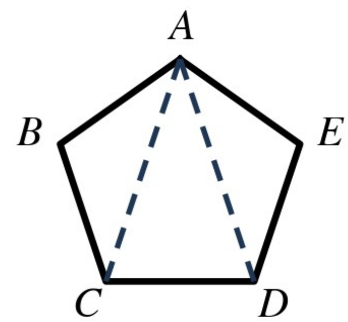
## **S**peak **L**ike **A** **M**athematician:

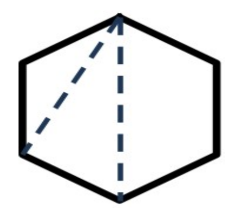
polygon:

side · vertex · diagonal:

equilateral · equiangular · regular:

convex · concave:



## Polygon Angle Theorems

**Polygon Interior Angles Theorem: the sum of the interior angle measures of a convex n-gon is ( ________ − 2) · ________ °.**

**Polygon Exterior Angles Theorem: the sum of the exterior angle measures of a convex polygon, one at each vertex, is ________ °.**

## Example 1 – Name That Polygon

Fill in the name of each polygon by its number of sides.

| # of sides | Name | # of sides | Name |
|---|---|---|---|
| 3 | | 8 | |
| 4 | | 9 | |
| 5 | | 10 | |
| 6 | | 12 | |
| 7 | | n | |

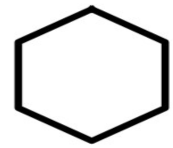
## Example 2 – Polygon or Not?

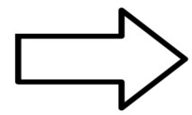
Tell whether each figure is a polygon. If it is, name it by its number of sides.

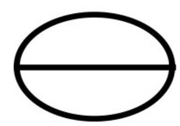
a) Figures 1 and 2 at the right.

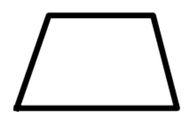
b) Figures 3 and 4 at the right.

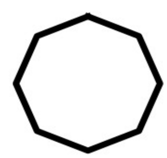
c) Figures 5 and 6 at the right.











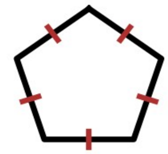
## Example 3 – Classify Each Polygon

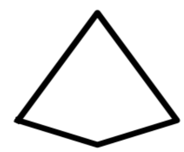
Tell whether each polygon is regular or irregular, and concave or convex.

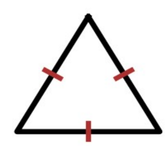
a) Figures 1 and 2 at the right.

b) Figures 3 and 4 at the right.

c) Figures 5 and 6 at the right.







## Example 4 – Interior Angle Sums

a) Find the sum of the interior angle measures of a convex nonagon.

b) Find the measure of each interior angle of a regular 20-gon.

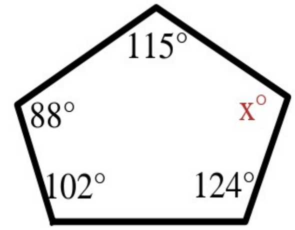
c) Find the sum of the interior angle measures of a convex 14-gon.

## Example 5 – Find the Value of x

The pentagon at the right has interior angles of 102°, 124°, x°, 115°, and 88°.

a) What must the five angles add to? Write the equation.

b) Solve for x.

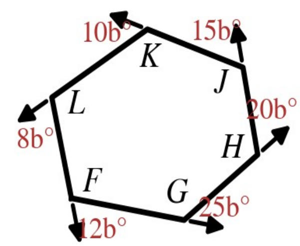



## Example 6 – Exterior Angles: Find b

In hexagon FGHJKL at the right, the exterior angles measure 12b°, 25b°, 20b°, 15b°, 10b°, and 8b°.

a) Write the equation and solve for b.

b) Find the largest exterior angle.




## Example 7 – Exterior Angles: Find r

In quadrilateral JKLM at the right, the exterior angles measure 3r°, 8r°, 5r°, and 4r°.

a) Find r.

b) Find each exterior angle. Check that they total 360°.